Test Environment

Framework: OP-TEE (Open Portable Trusted Execution Environment)
Test Suite: xtest benchmark (level 15)
Test Cases: 7 benchmarks, 24 subtests
Result: All tests passed (0 failures)

Fastest Read Speed
1000 kB/s
TEE Secure Storage (1KB chunks)

Best Write Speed
32.8 kB/s
TEE Secure Storage (2KB chunks)

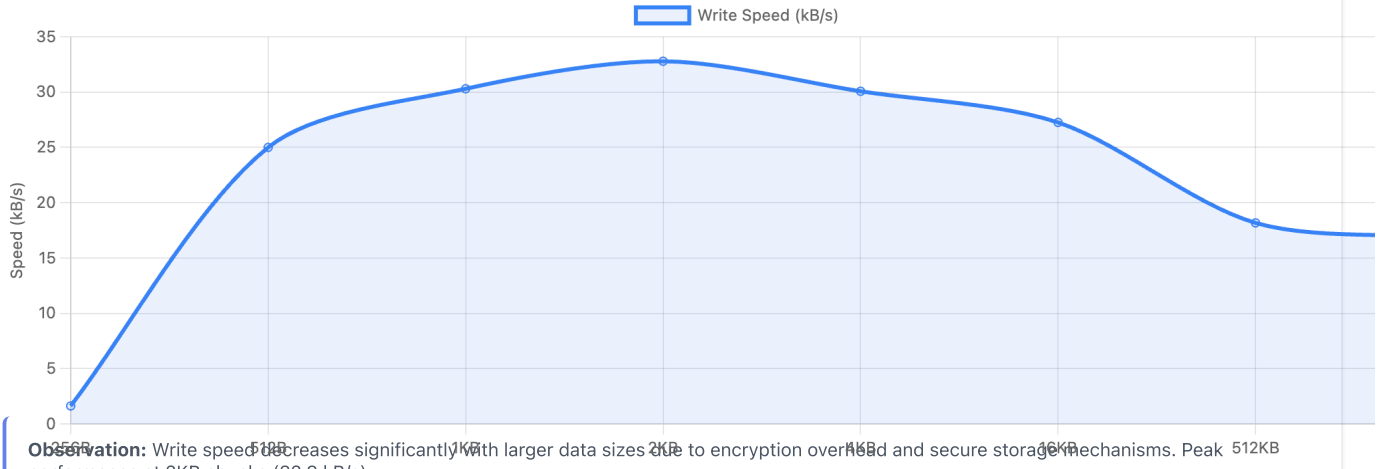
Crypto Performance
7.92 MB/s
SHA-226 Hashing

AES Encryption
3.32 MB/s
AES-ECB Mode

Performance Note

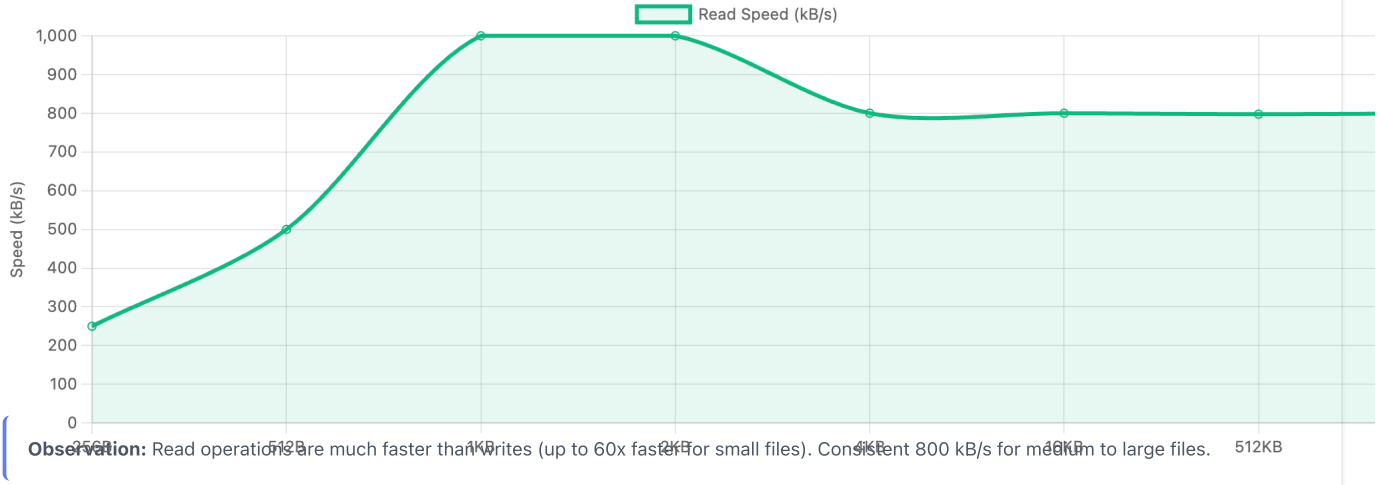
TEE operations are intentionally slower than normal world operations due to security overhead (context switching, encryption, isolation). This is expected and necessary for security guarantees.

TEE Trusted Storage - WRITE Performance



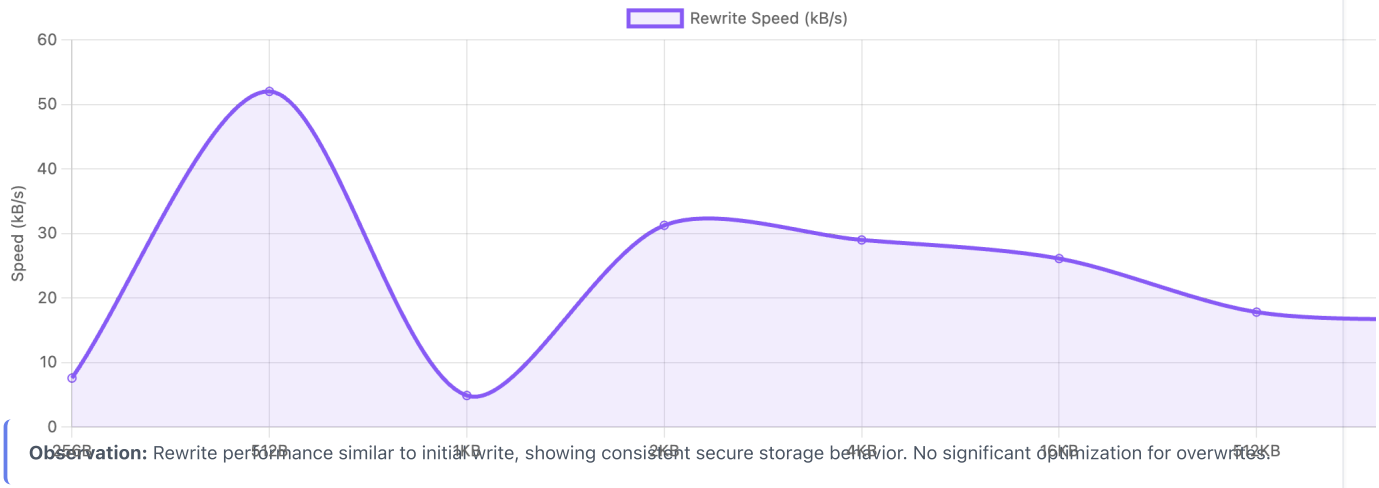
Observation: Write speed decreases significantly with larger data sizes due to encryption overhead and secure storage mechanisms. Peak performance at 2KB chunks (32.8 kB/s).

TEE Trusted Storage - READ Performance



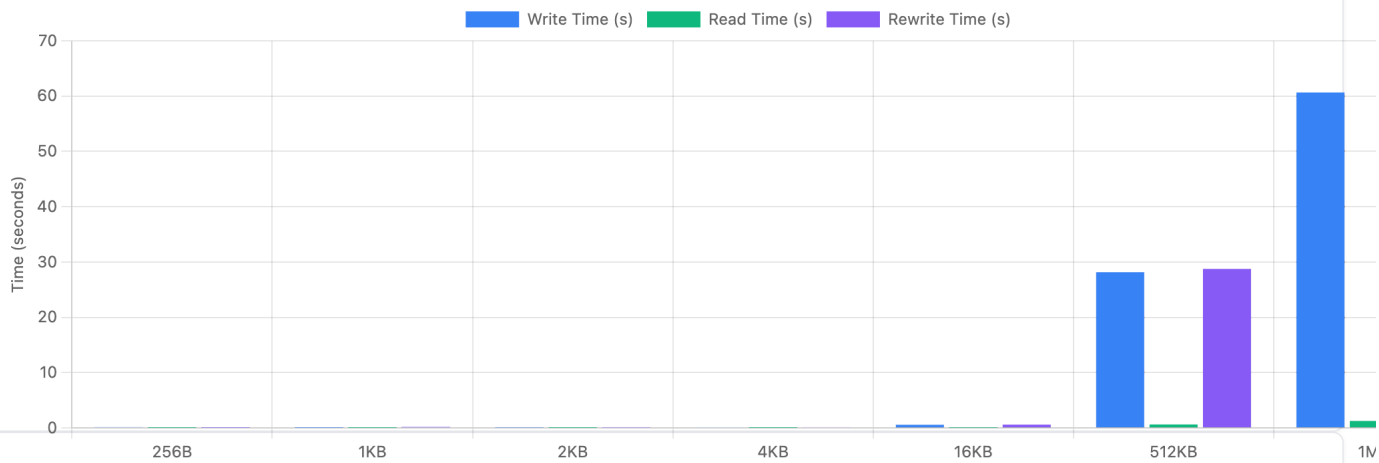
Observation: Read operations are much faster than writes (up to 60x faster for small files). Consistent 800 kB/s for medium to large files.

TEE Trusted Storage - REWRITE Performance

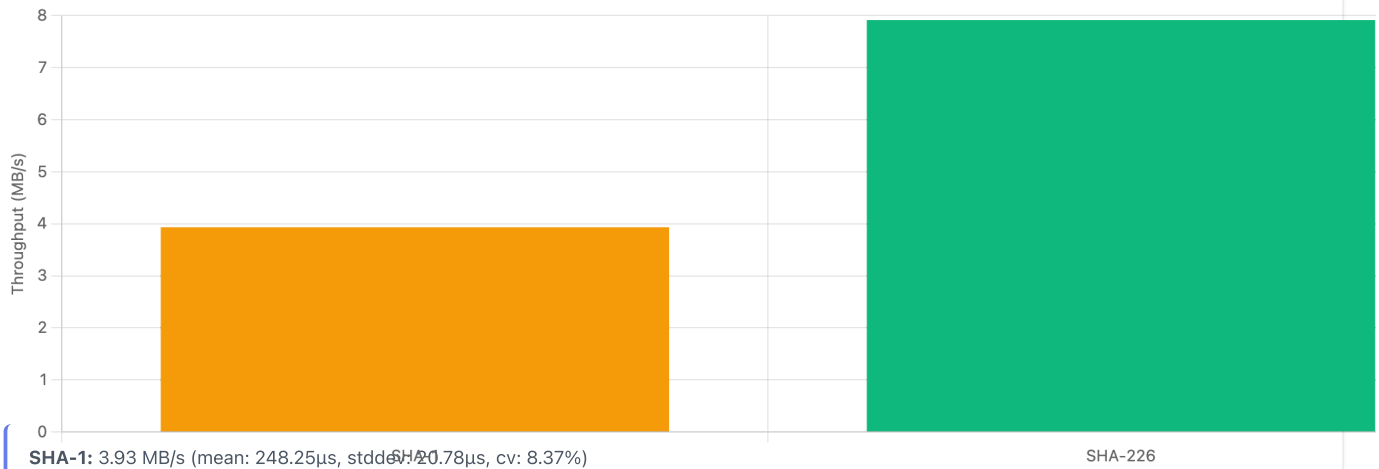


Observation: Rewrite performance similar to initial write, showing consistent secure storage behavior. No significant optimization for overwrites.

Operation Time Comparison

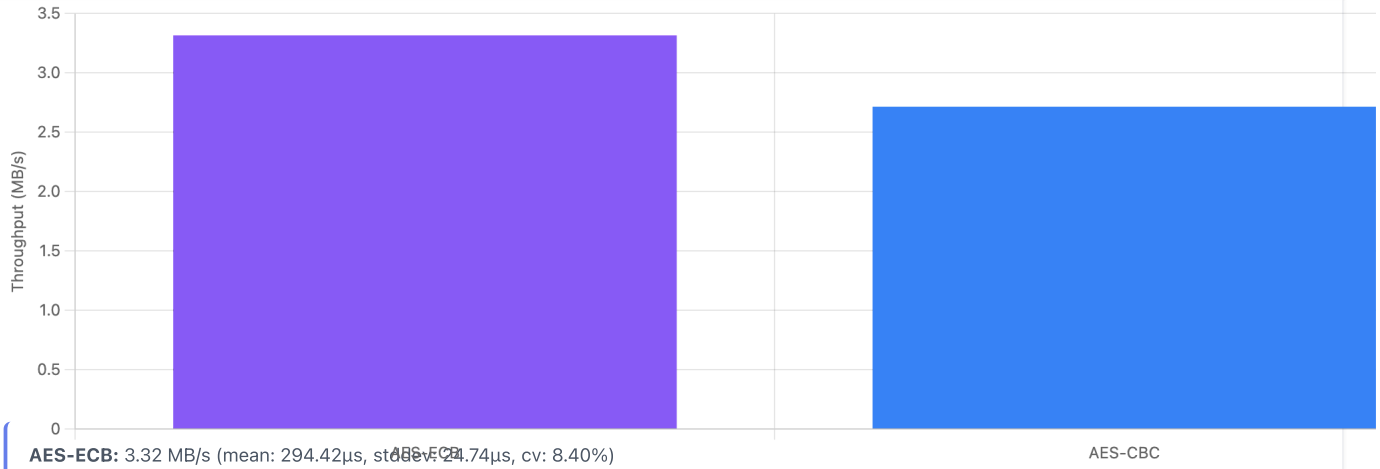


SHA Hashing Performance



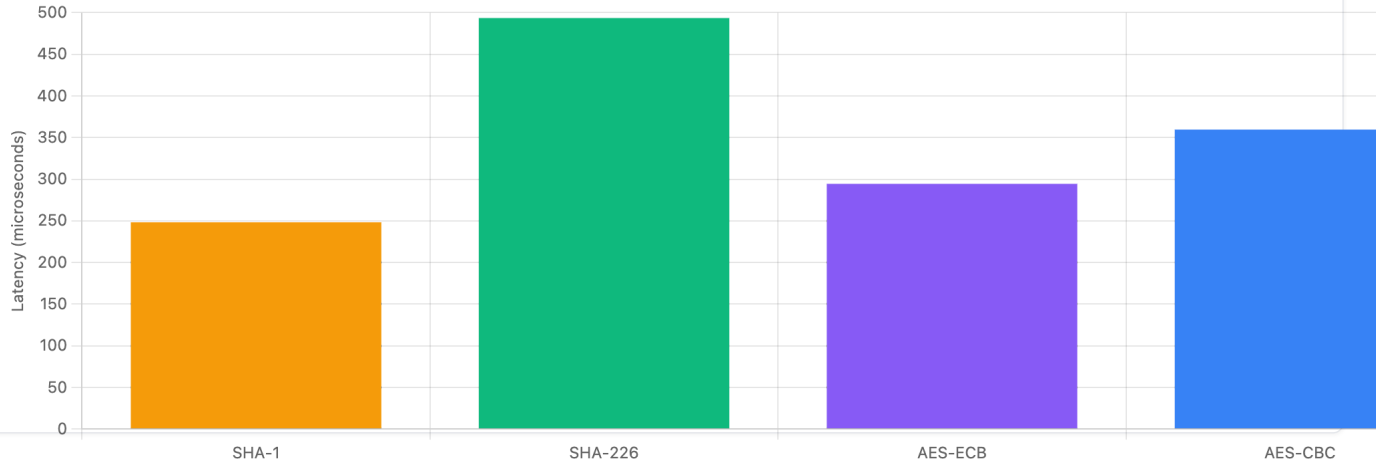
SHA-1: 3.93 MB/s (mean: 248.25µs, stddev: 20.78µs, cv: 8.37%)
SHA-226: 7.92 MB/s (mean: 493.51µs, stddev: 17.91µs, cv: 3.63%)
SHA-226 provides better throughput with lower variance, indicating more stable performance.

AES Encryption Performance



AES-ECB: 3.32 MB/s (mean: 294.42µs, stddev: 24.74µs, cv: 8.40%)
AES-CBC: 2.72 MB/s (mean: 359.58µs, stddev: 20.90µs, cv: 5.81%)
ECB mode is faster but CBC provides better security. CBC has lower variance.

Cryptographic Operation Latency



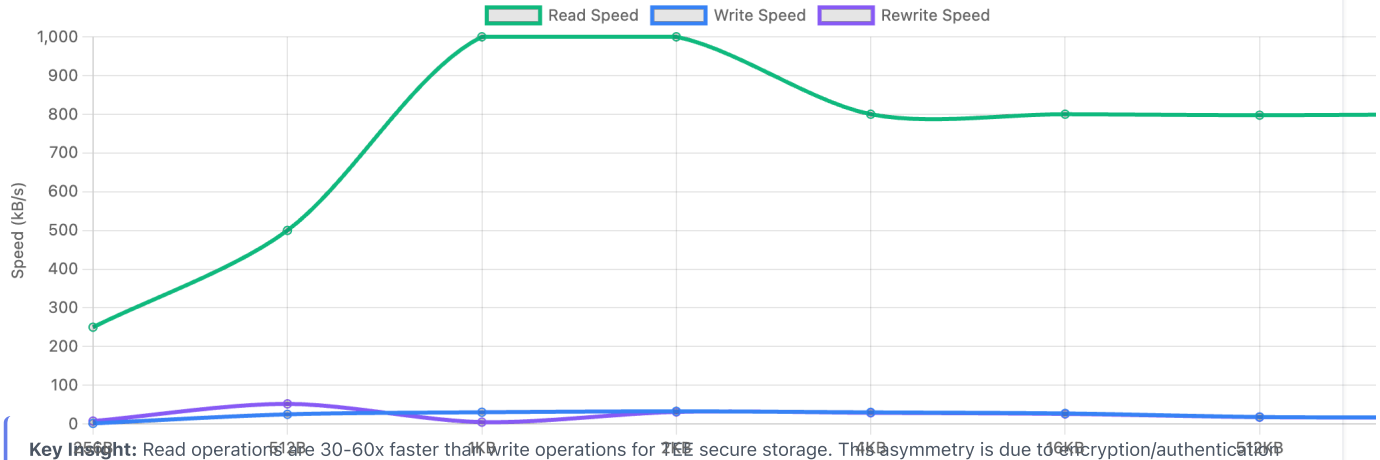
SHA-1 Throughput
3.93 MB/s
Mean: 248.25µs | CV: 8.37%

SHA-226 Throughput
7.92 MB/s
Mean: 493.51µs | CV: 3.63%

AES-ECB Throughput
3.32 MB/s
Mean: 294.42µs | CV: 8.40%

AES-CBC Throughput
2.72 MB/s
Mean: 359.58µs | CV: 5.81%

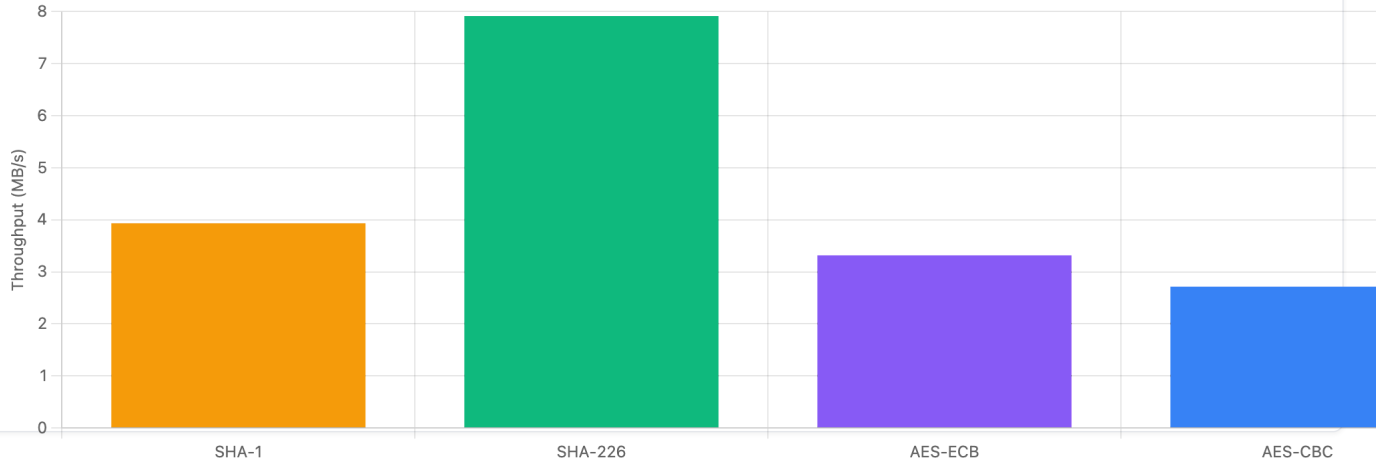
Read vs Write vs Rewrite Performance



Key Insight: Read operations are 30-60x faster than write operations for TEE secure storage. This asymmetry is due to encryption/authentication

overhead during writes.

Cryptographic Algorithm Comparison



Performance Characteristics Summary

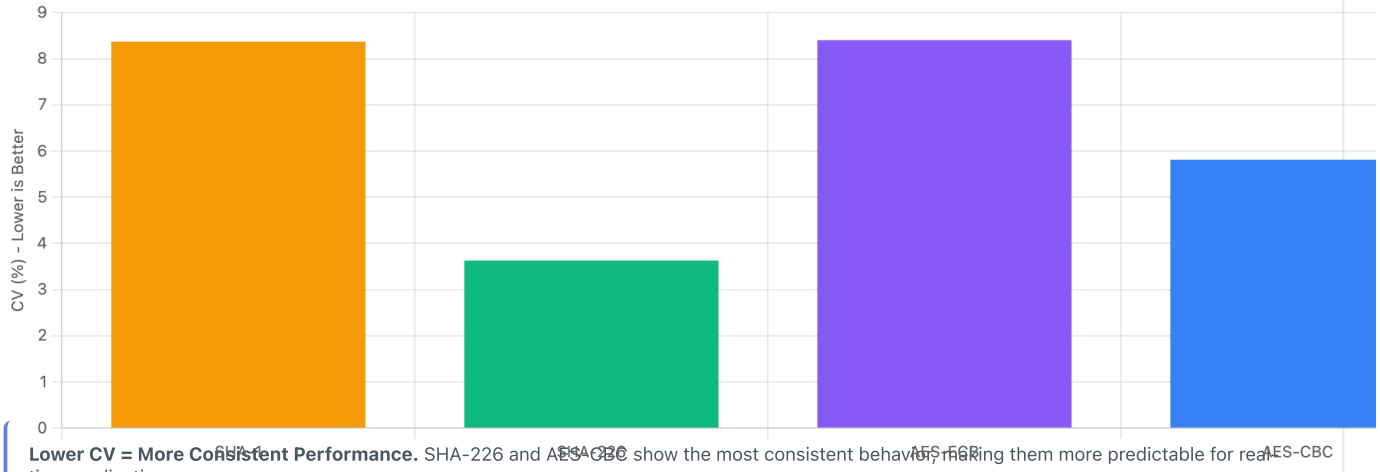
Storage Operations:

- Write: 1.6-32.8 kB/s (size dependent)
- Read: 250-1000 kB/s (much faster)
- Rewrite: Similar to write performance

Cryptographic Operations:

- SHA-226 outperforms SHA-1 (7.92 vs 3.93 MB/s)
- AES-ECB faster than CBC but less secure
- All operations show good consistency (CV < 10%)

Performance Variability (Coefficient of Variation)



Lower CV = More Consistent Performance. SHA-226 and AES-CBC show the most consistent behavior, making them more predictable for real-time applications.